Customer – Transactions Activity & Spending Patterns

DS 5500 : EDA

18th February 2026

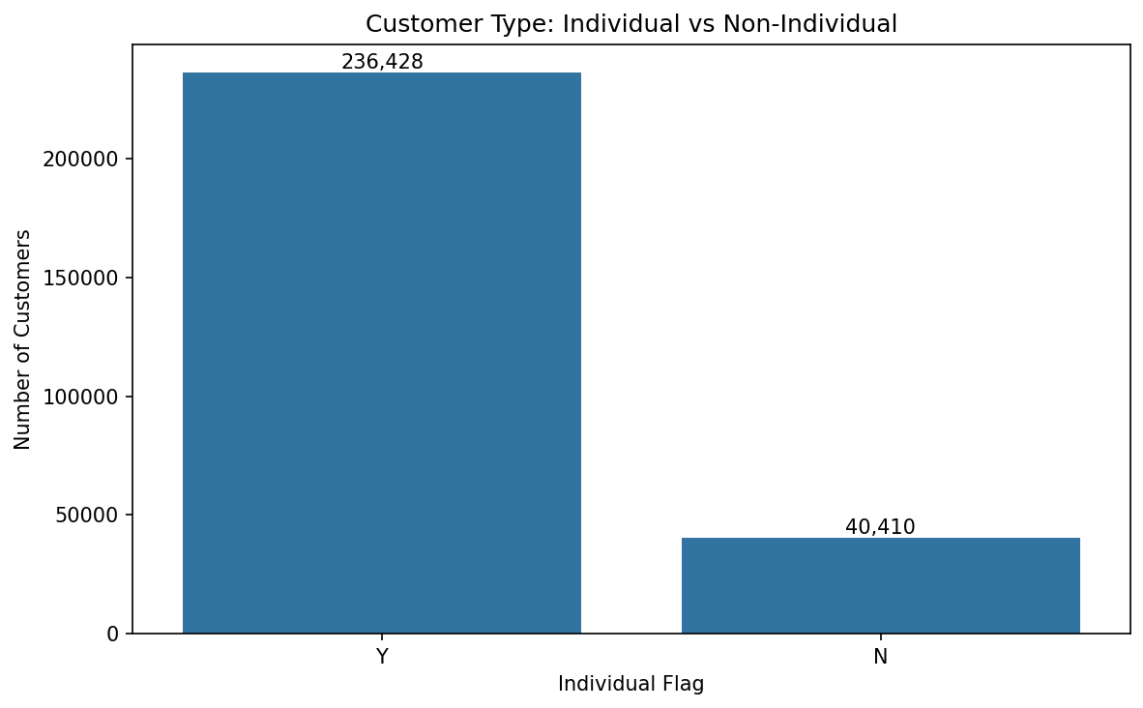
Proposal -

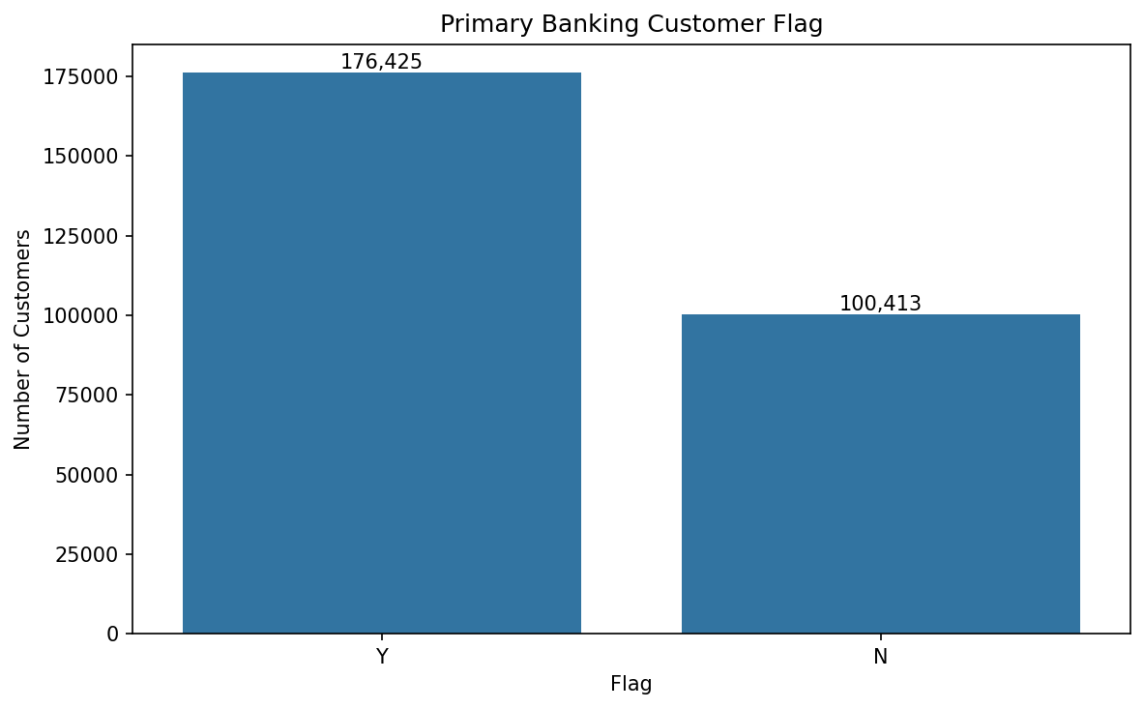
Bank transactions record the rhythm of daily life and though masked and concise, they reveal a customer's habits, routines, and financial behavior. Analyzing these patterns allows a bank to understand who their customers truly are, not just demographically, but behaviorally.

The initial goal of this project was fraud detection and anomaly analysis. However, early exploration revealed that making supervised fraud detection is not feasible. This led to a pivot toward customer segmentation and spending pattern prediction, a direction that provides measurable outcomes and concrete business value.

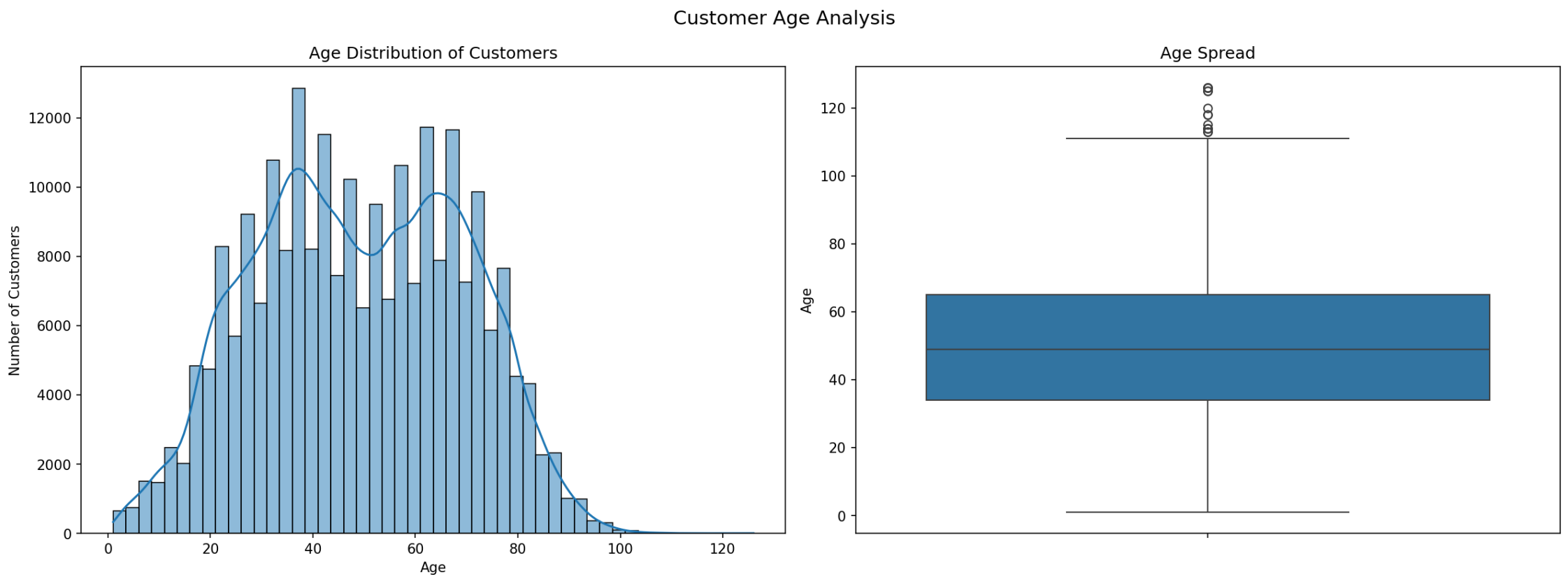
The project now aims to build a **customer intelligence system** that segments Bangor Savings Bank customers by their transaction behavior and predicts spending pattern changes. This enables the bank to **identify customers who may be struggling financially** before their situation worsens, determine which customers are **good candidates for the bank's rewards program** (Buoy Local), deliver **personalized cashback incentives** aligned with actual spending habits, and **promote local businesses** to relevant customer segments. This system can complement the bank's existing recommendation engine with data-driven behavioral insights.

Customer Types



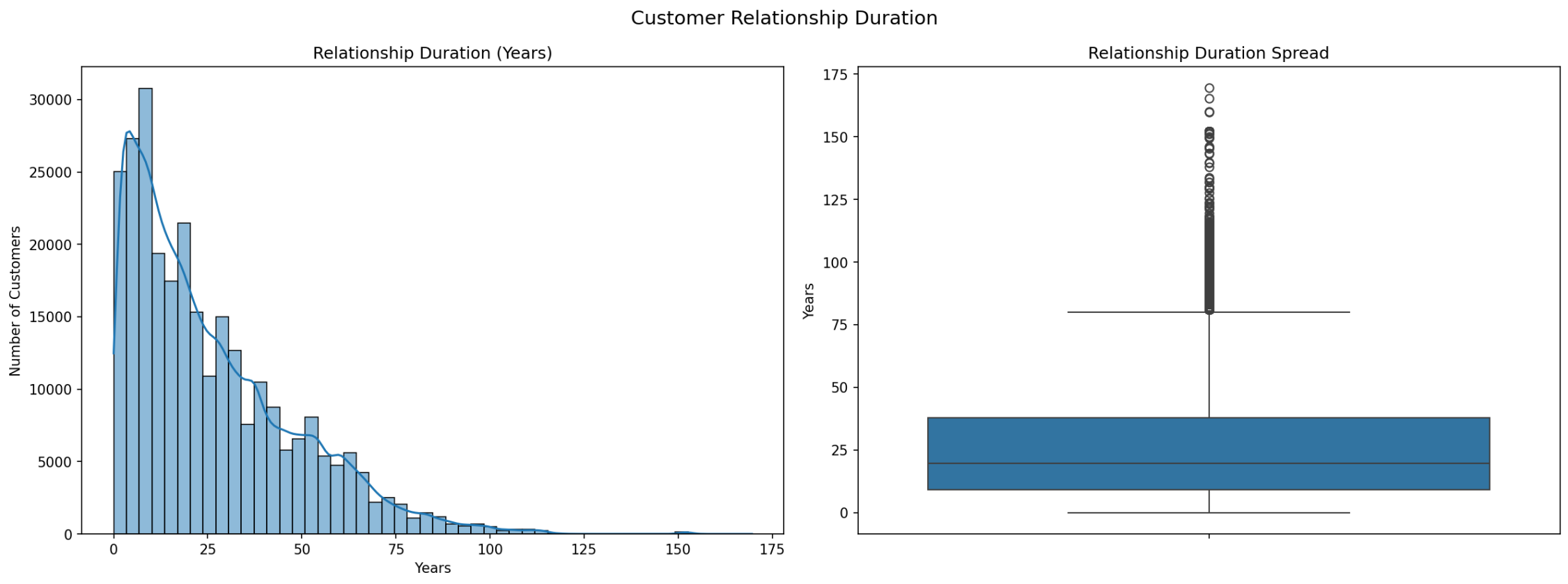


Customer Age

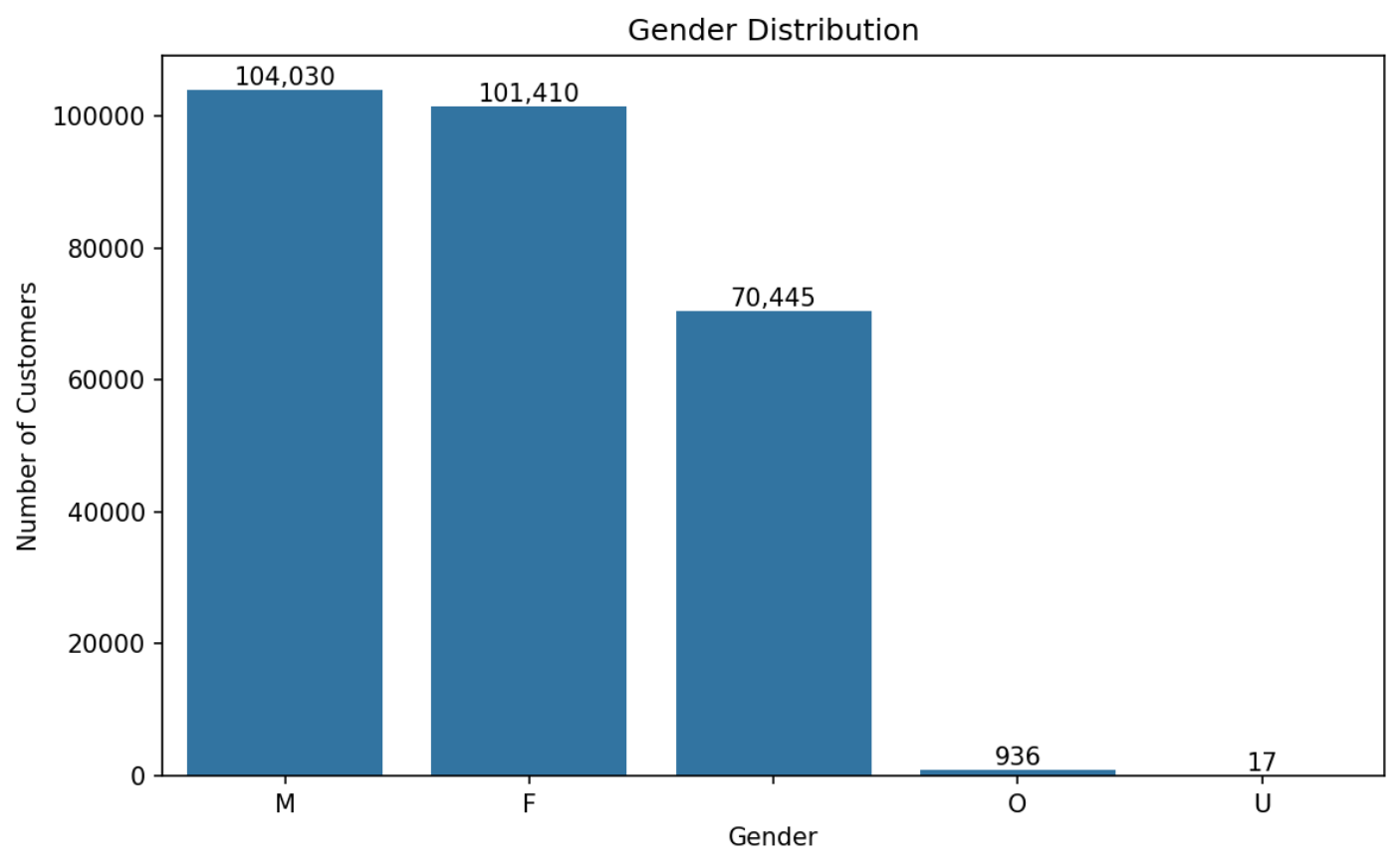


40k customers from 278k customers (~14%) with age 0 were excluded for this plot

Customer Relationship

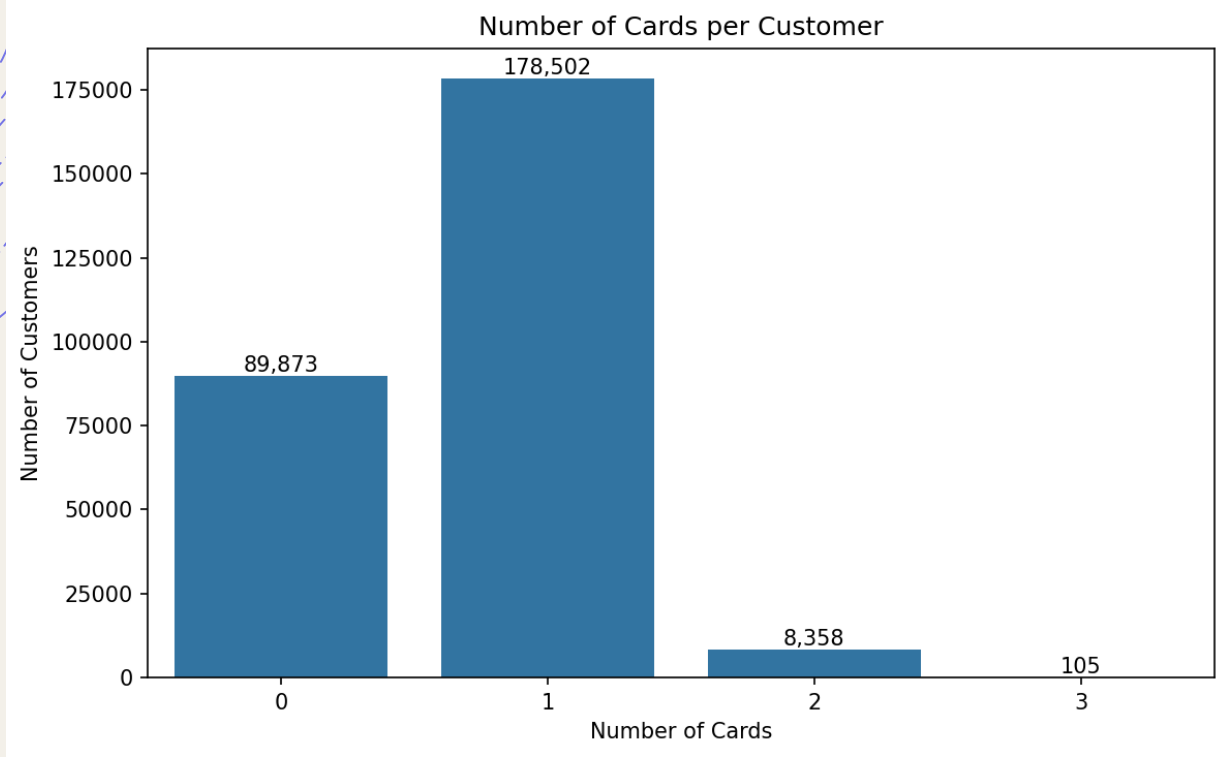


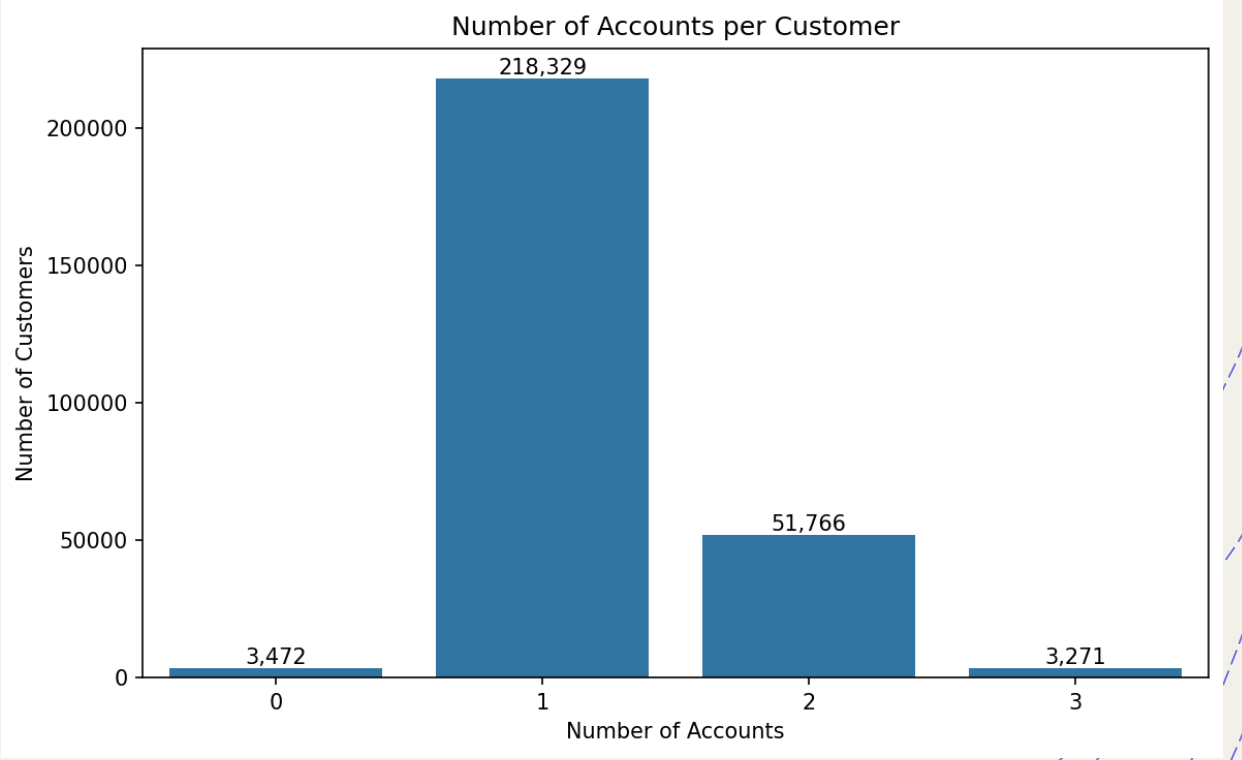
Customer Genders



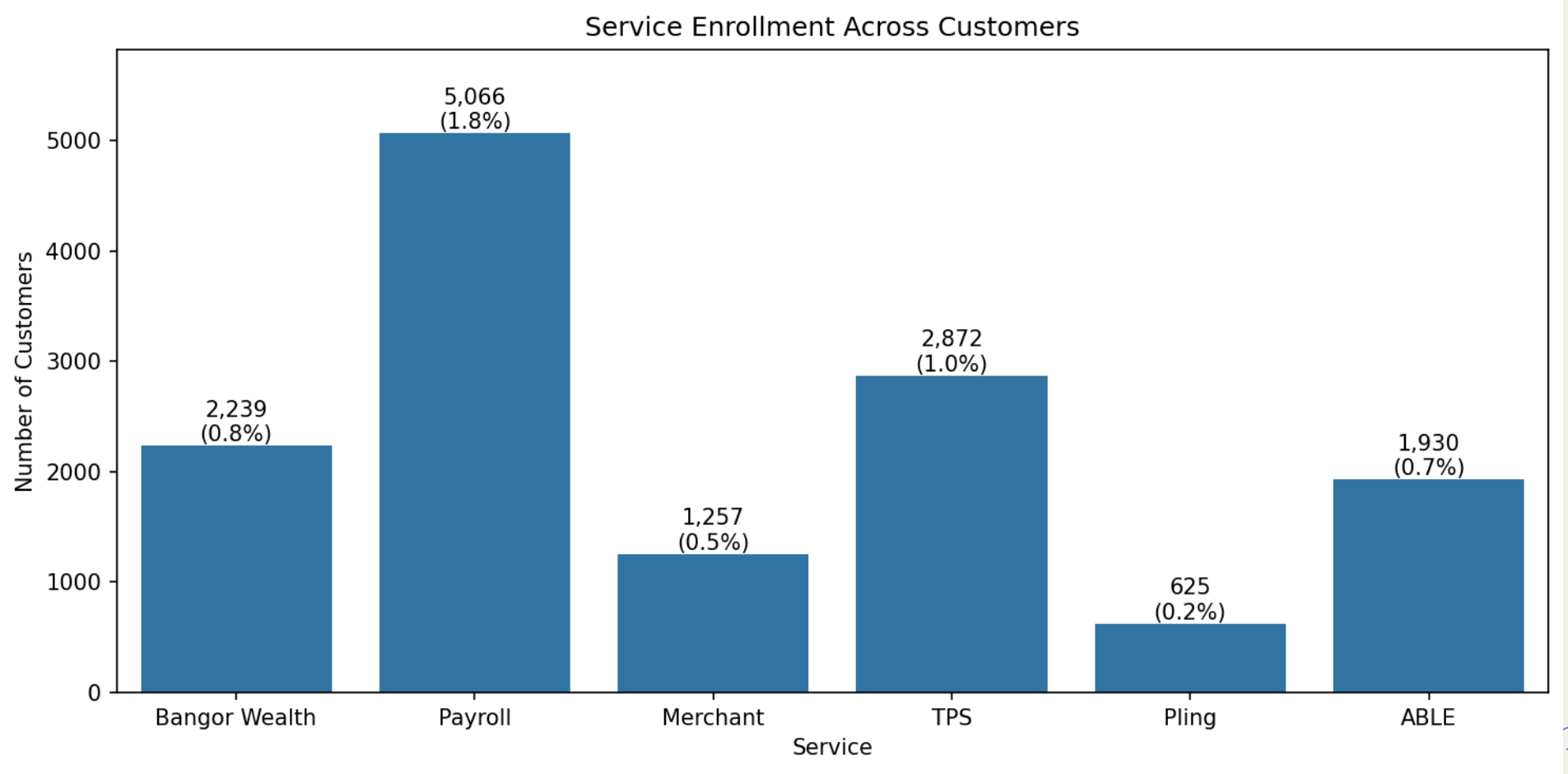
While creating features, plan is to have 3 parameters – isMale, isFemale, isOther (includes blank, other and undeclared)

Customer Holdings

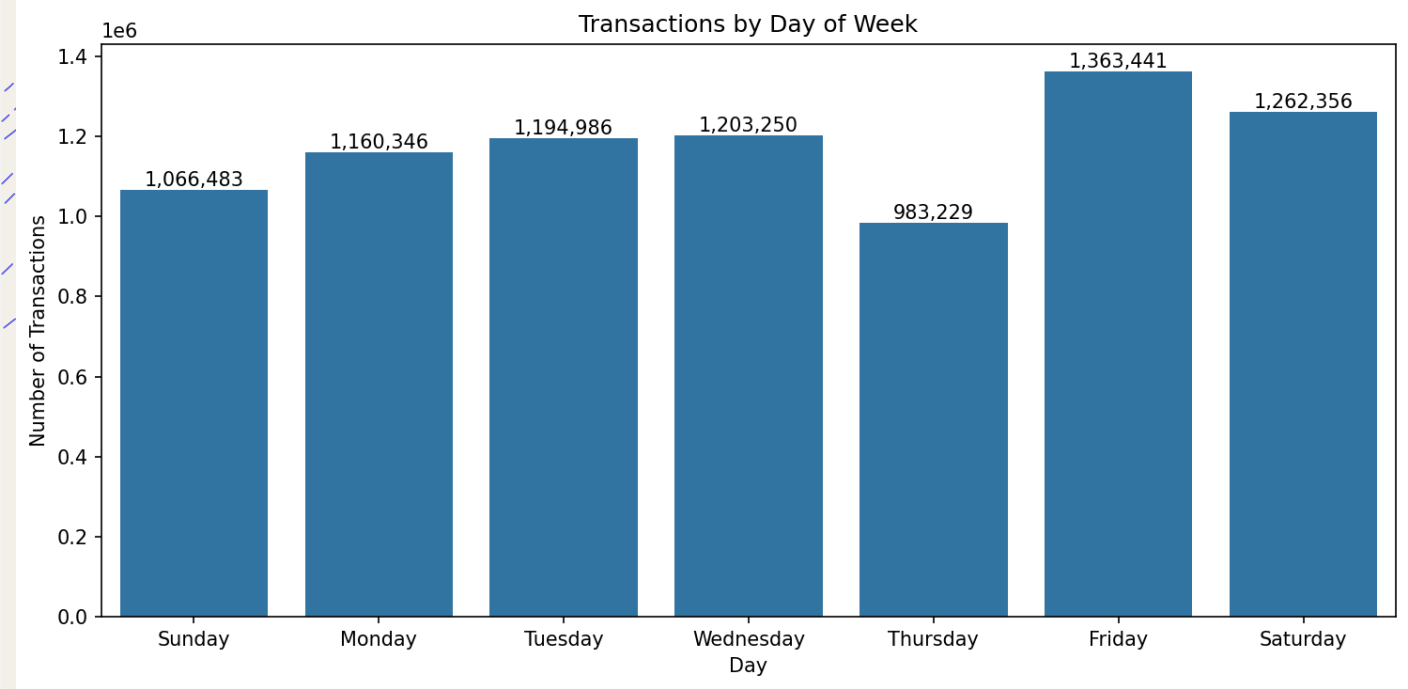


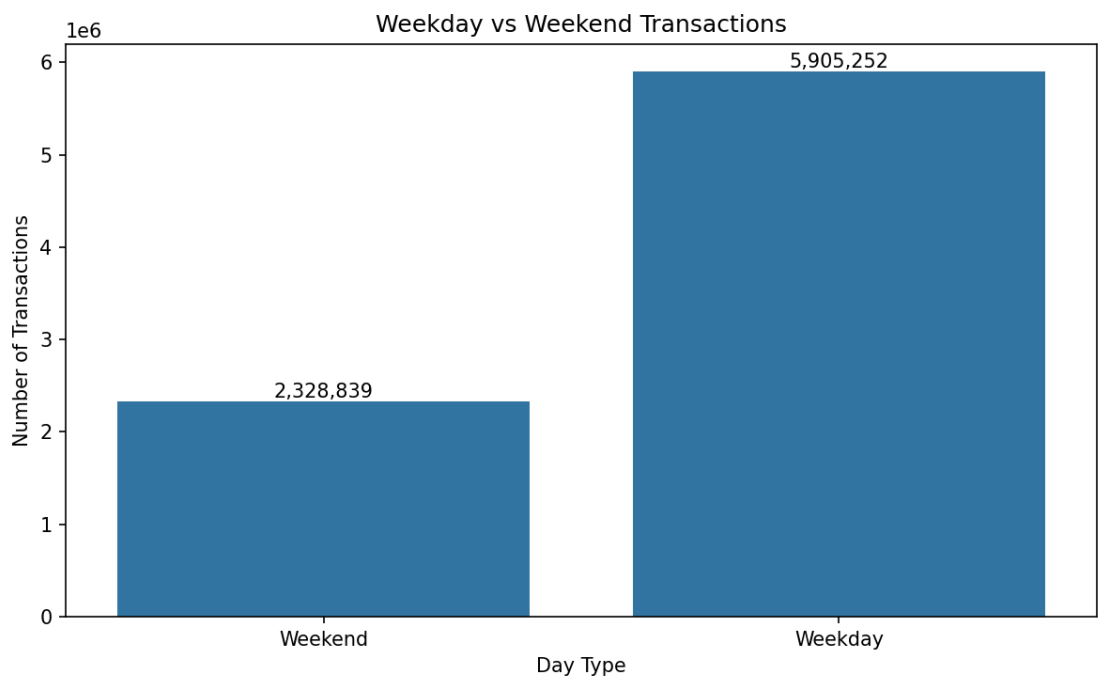


Customer Services

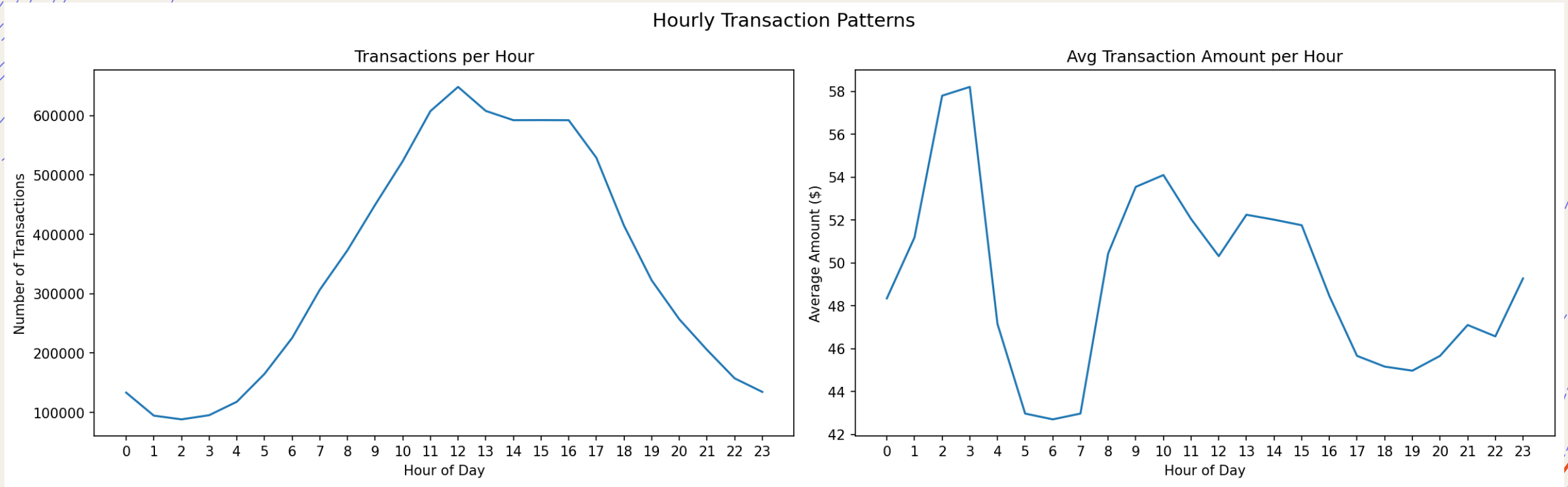


Transactions per day

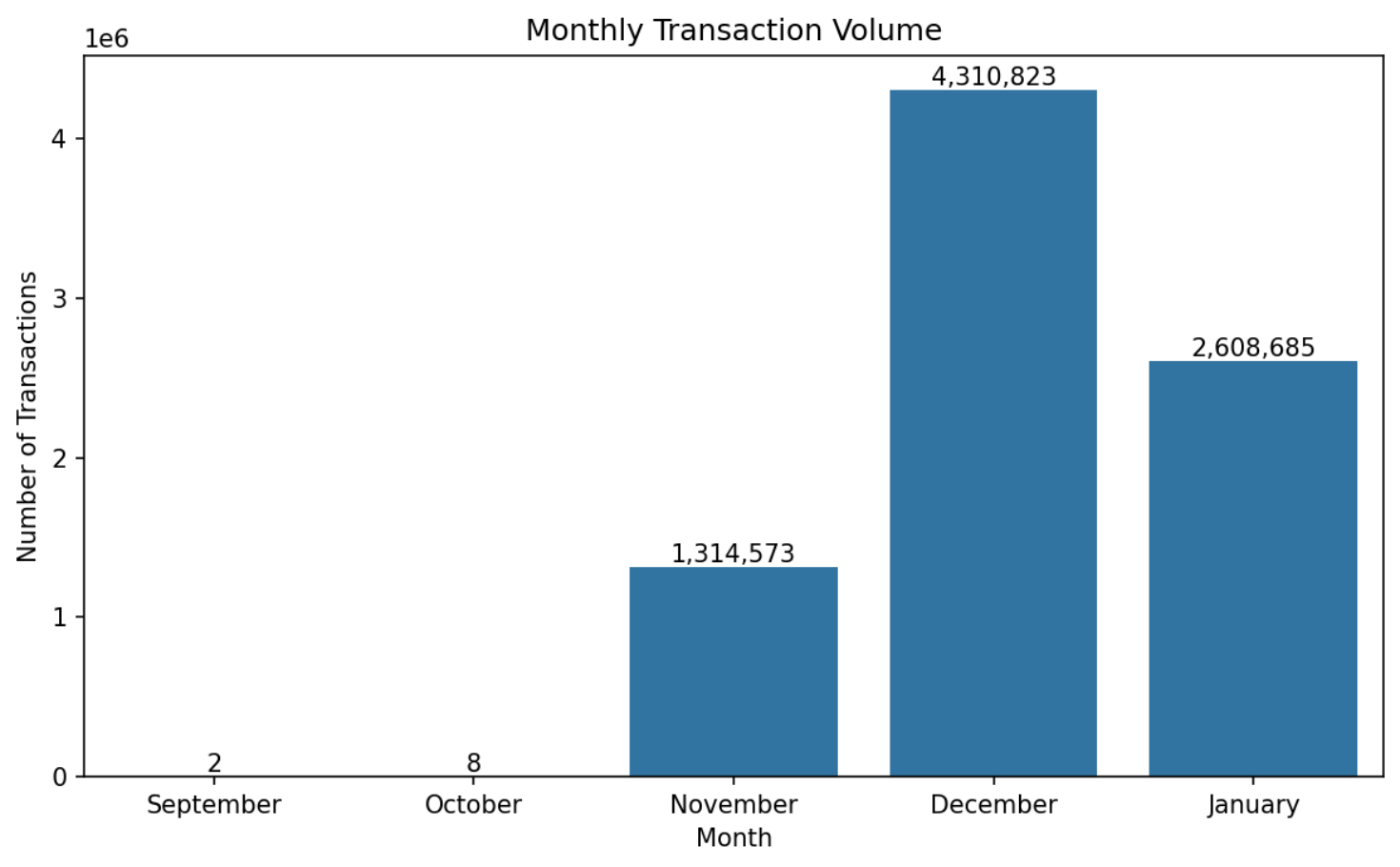




Transactions and avg amount per hour

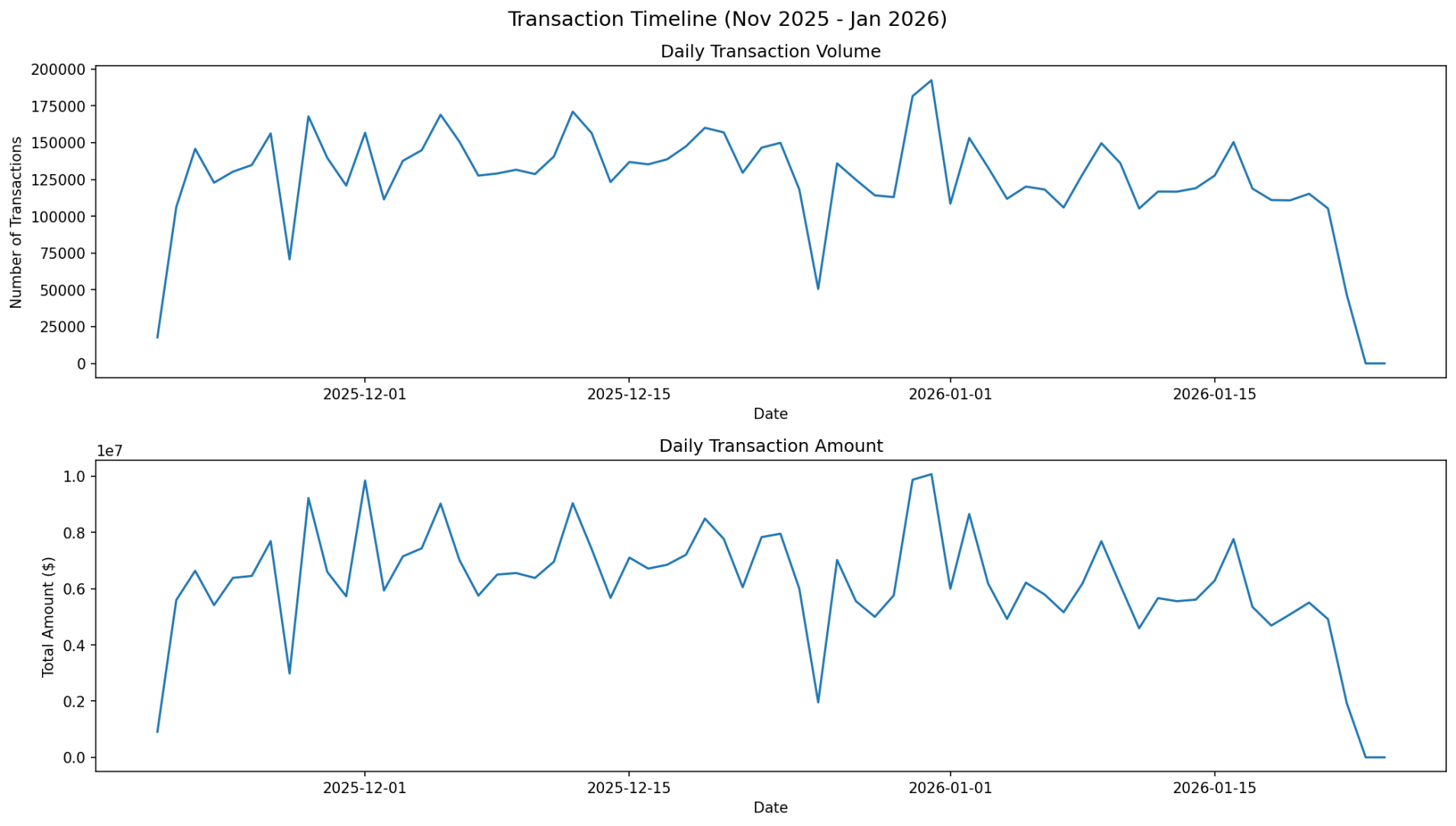


Transactions monthly

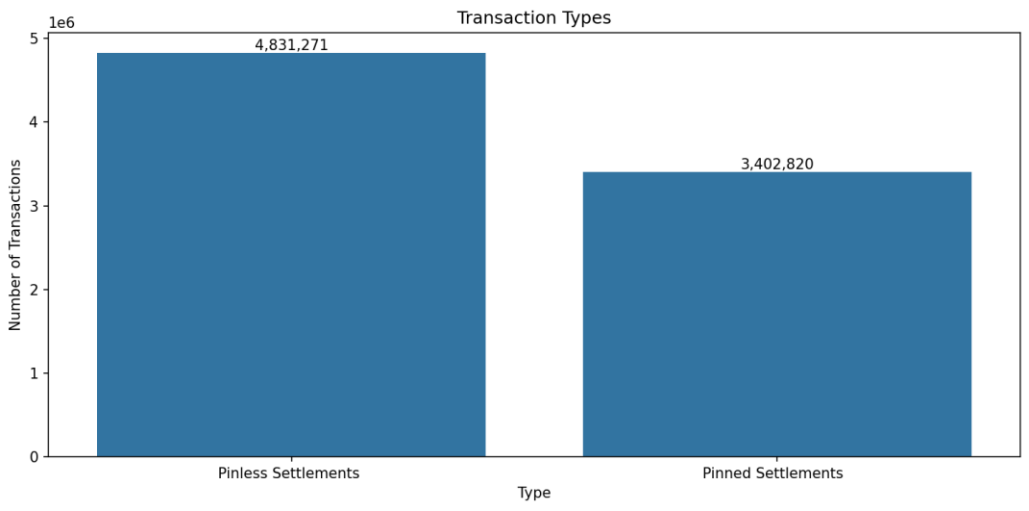


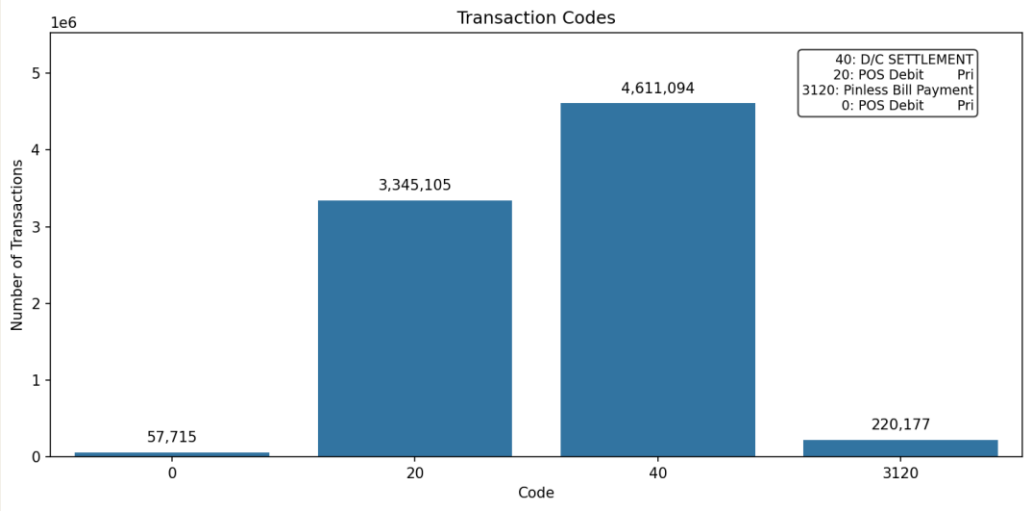
November data begins around the 21st, representing approximately 10 days of transactions. December, with a full 31 days, has roughly 3x the transaction volume,almost consistent with that. Data ends January 24th.

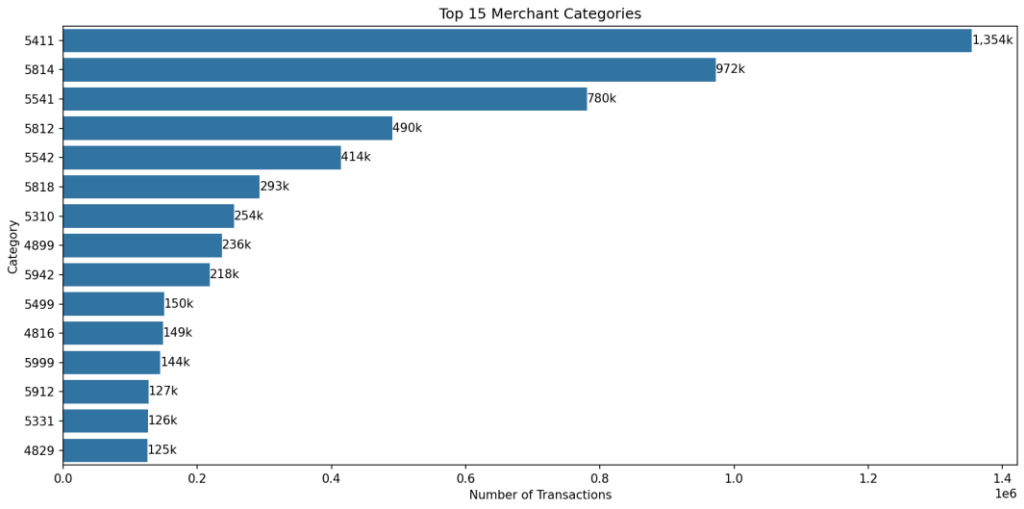
Transactions timeline

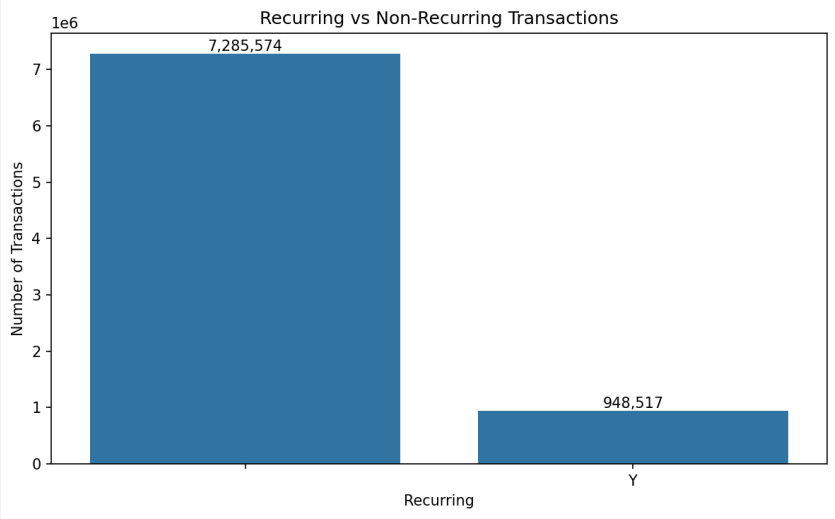


Transaction types

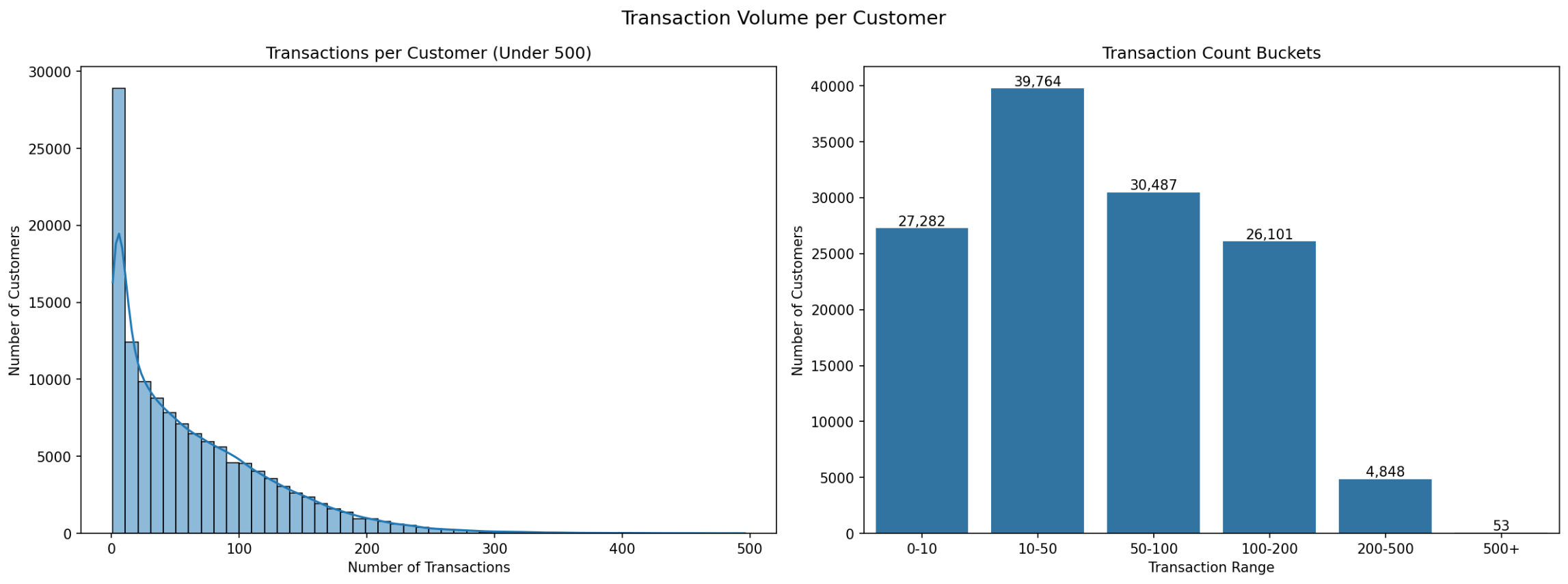
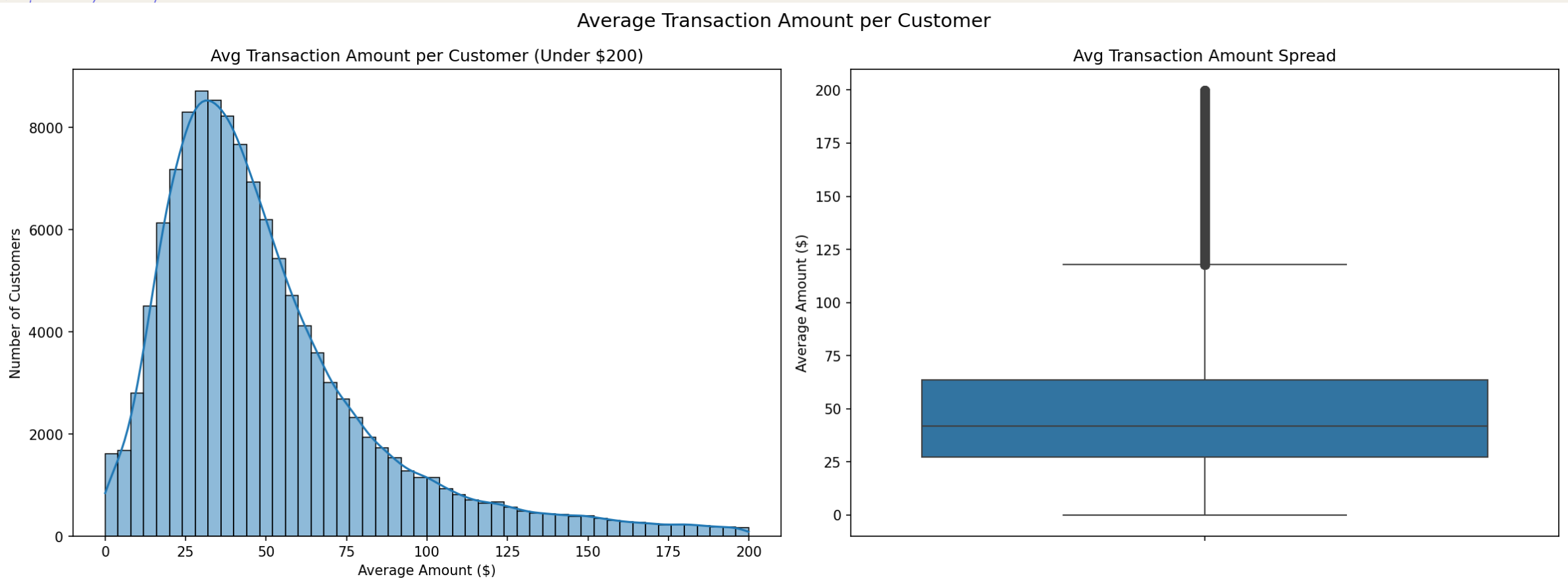








Transactions per customers (a)



Transactions per customers -avg (b)